Prativimb

A patient twin that answers what if, not just what next.

Hybrid digital twin for Type 2 diabetes and cardiovascular risk, built for Indian patients.

Happiest Health hackathon. Digital Twin track, Phase 1: prototype and code.

Team pushkalayeddula, Anurag University

Built by Pushkala Yeddula: data, twin, models, dashboard, tests and documents

Prativimb is Sanskrit for reflection. Patient data shown is synthetic; real data validation uses open access CGMacros.

WHY NOW

The problem

- India has about 101 million adults with diabetes and 136 million with prediabetes (ICMR-INDIAB, Lancet Diabetes & Endocrinology, 2023).
- Most see a doctor every three months. Glucose, sleep and heart rhythm change every five minutes.
- CGMs and watches now record that stream, but they show what already happened.
- Risk scores say who is at risk. They do not say what to change today, or how sure they are.

What a clinician needs at the bedside

- Which patients will go out of range in the next two hours?
- Why, in words, not feature names?
- What would help this patient: a walk, a different grain, an earlier dinner?
- Who is drifting towards a cardiac event this week?

OUR ANGLE

Why a twin, and not another classifier

	Risk classifier	Prativimb twin
Question	What happens next?	What happens next, and what if we change something?
Model	Learned from data only	Physiology (ODE) plus a learned residual
Personal	Population average	Parameters fitted to each patient's first week
Out of range inputs	Unreliable	Physics extrapolates plausibly
Uncertainty	A score	Calibrated interval with stated coverage
Reasons	Feature importances	mg/dL contributions, one plain sentence

Prativimb is a reflection of the patient you can question.

METHOD

Our approach in four ideas

1. Hybrid twin

Bergman minimal model extended with meals, exercise, sleep debt, circadian rhythm, night shifts and metformin. EHR sets the prior, the first week of wearables fits 8 personal parameters, LightGBM learns what the physics misses.

2. Counterfactuals

Re-simulate the next six hours after a walk, a millet swap, a shifted dinner, a missed dose, more sleep or a night shift. Shows peak and time above range for both.

3. Honest uncertainty

Conformalised quantile regression gives 90% intervals that reach 89.2% coverage at 60 min on unseen patients. Alerts fire only when the lower bound crosses 180.

4. India first

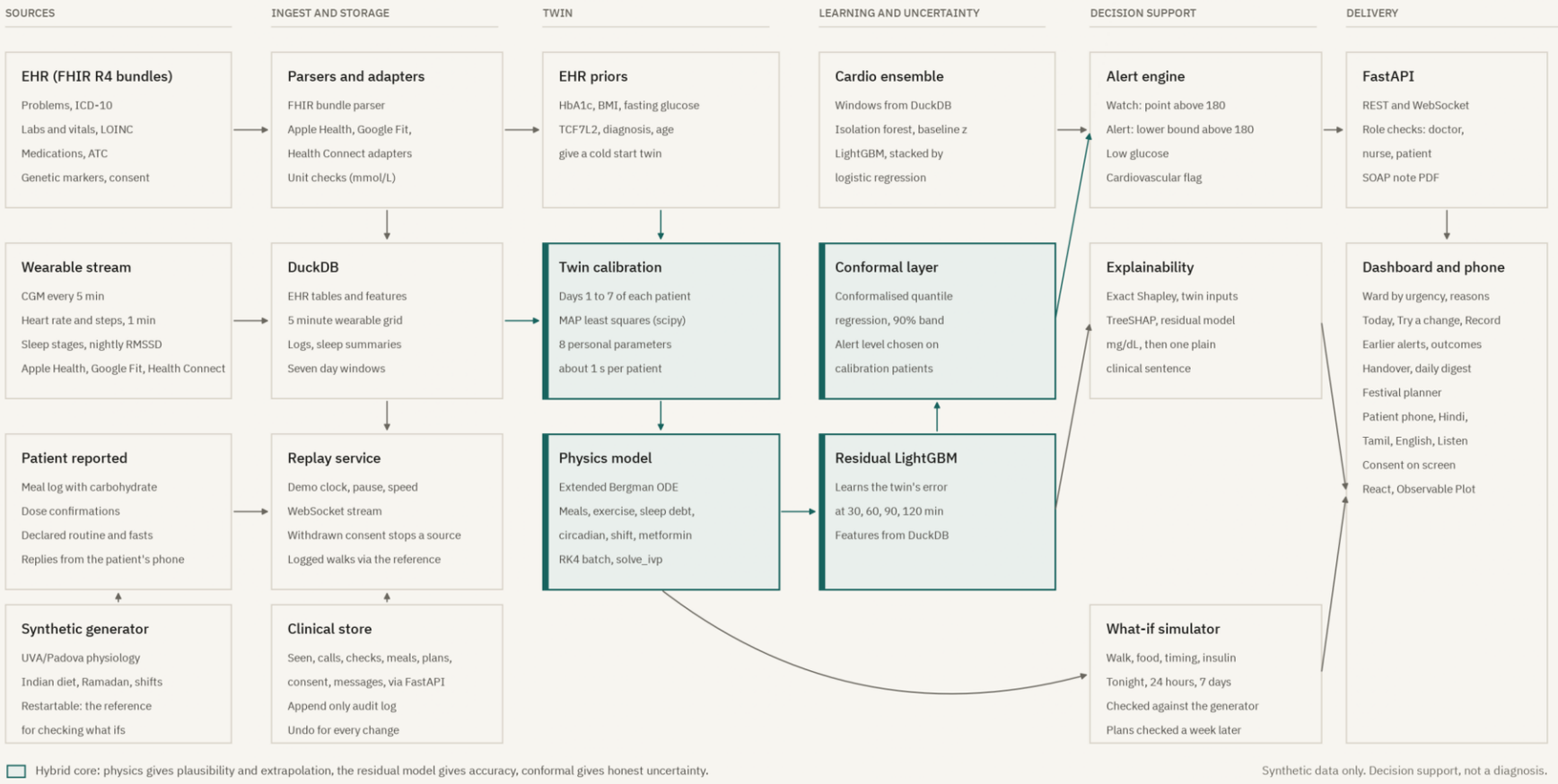
Rice, roti, idli and iftar meals with their absorption speeds; Ramadan, Shivaratri, Ekadashi and Holi in the calendar; night shift workers; cardiovascular risk at lower BMI.

SYSTEM

Architecture

Prativimb architecture

Hybrid patient twin: mechanistic physiology, learned residual, calibrated uncertainty, plain language reasons.

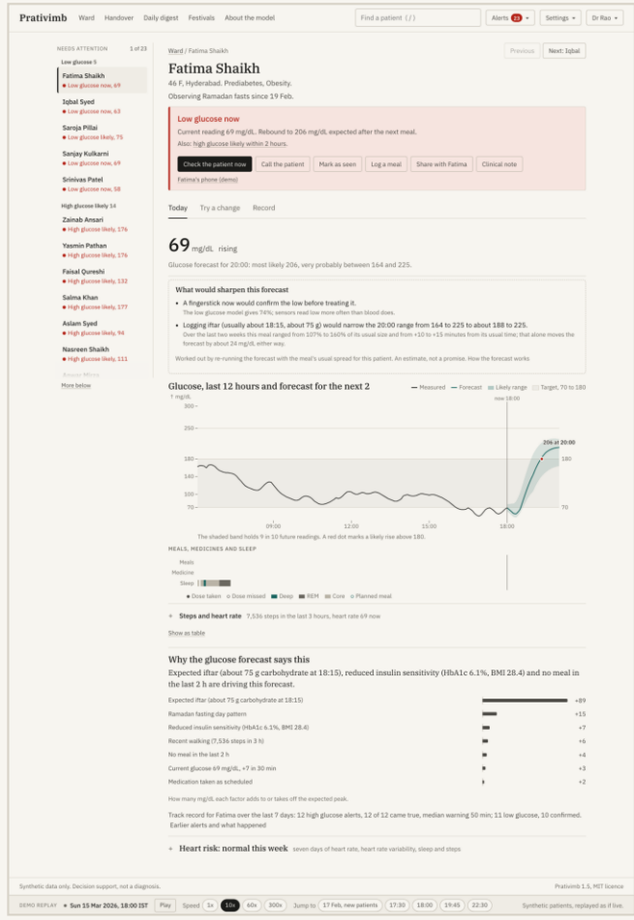


Hybrid core: physics gives plausibility and extrapolation, the residual model gives accuracy, conformal gives honest uncertainty.

Synthetic data only. Decision support, not a diagnosis.

Data: synthetic, Indian, reproducible

- 200 fictional patients, 30 days each (14 Feb to 15 Mar 2026): healthy, prediabetes and T2D; hypertension, dyslipidaemia, family history.
- EHR as FHIR R4 bundles: ICD-10, LOINC, ATC, TCF7L2, KCNQ1, SLC30A8 and 9p21 markers, DPDP style consent.
- Glucose from the UVA/Padova model with simglucose parameters and Dexcom noise, adapted for T2D. HR every minute, sleep stages, nightly HRV, steps.
- Apple Health and Google Fit export formats, parsed back by our adapters.
- One command regenerates everything in about 20 seconds from seed 20260315.



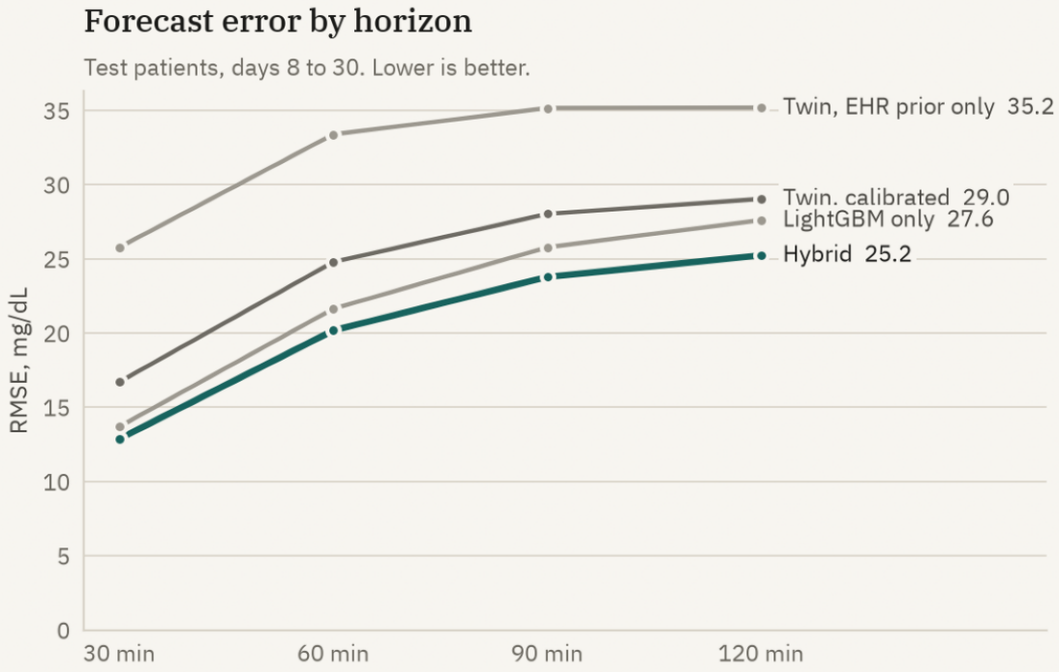
Fatima (synthetic): Ramadan pattern in her glucose profile.

METHOD

Models

Component	What it does
Twin ODE	dG/dt = -(SG + X)G + SG Gb H(t) + cf Ra(t) - eE E G, with insulin action, insulin, two gut compartments, exercise and metformin states
Calibration	MAP least squares on days 1 to 7 with EHR priors; about one second per patient
Residual model	LightGBM on twin error at 30, 60, 90, 120 min from CGM history, meals, activity, sleep, adherence, EHR
Uncertainty	CQR intervals at 90%; the alert tier uses the lower bound of the 50% interval, chosen on calibration patients
Cardio	Isolation forest and personal z scores (unsupervised) plus LightGBM, stacked by logistic regression
Explanations	Exact Shapley values over six twin inputs plus TreeSHAP on the residual, in mg/dL

Results: glucose forecasting

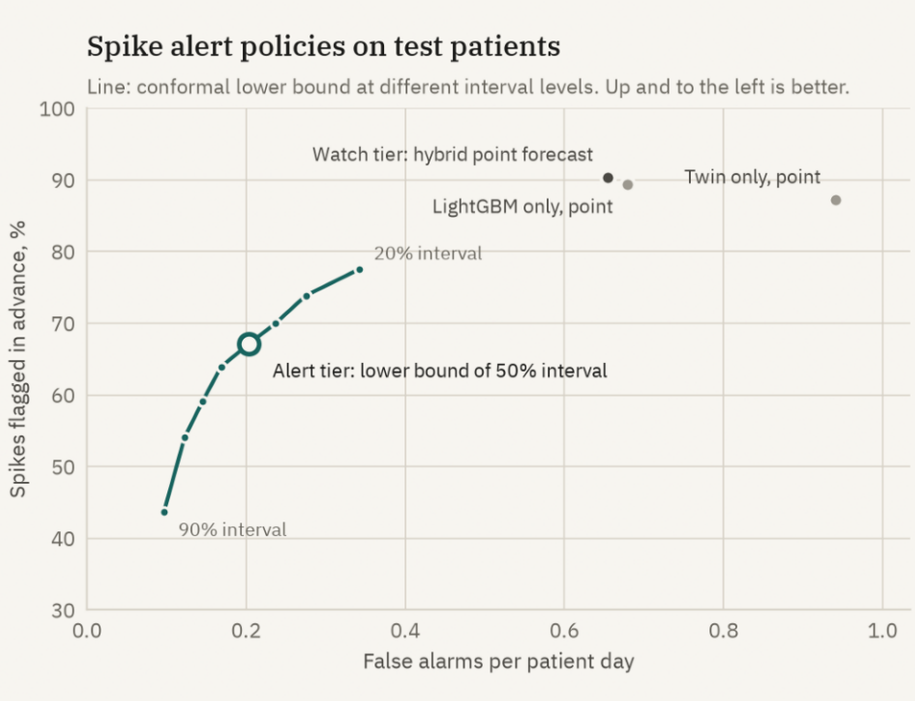


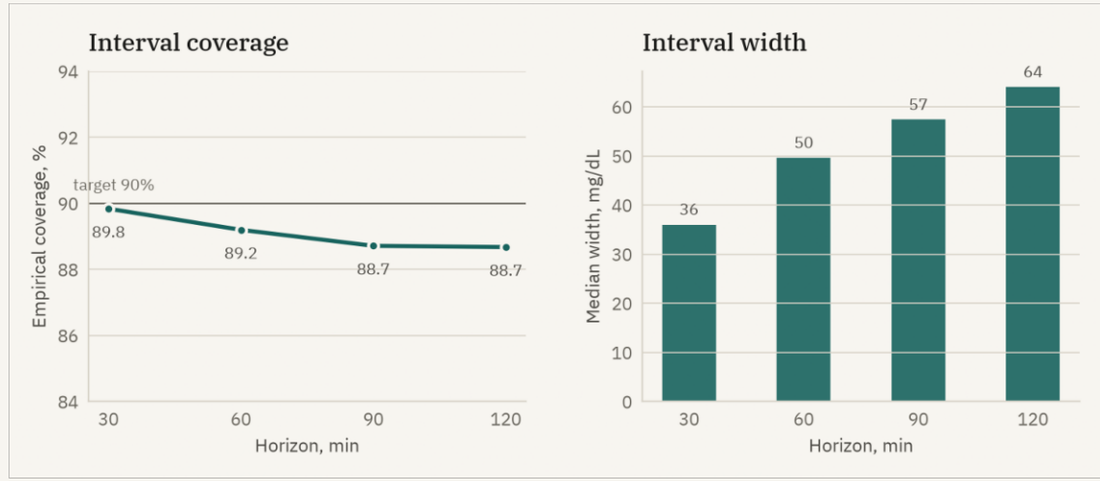
RMSE, mg/dL	30	60	120
Twin, EHR only	25.8	33.4	35.2
Twin, calibrated	16.7	24.8	29.0
LightGBM only	13.7	21.6	27.6
Hybrid	**12.9**	**20.2**	**25.2**

Hybrid beats both of its parts at every horizon. MARD at 60 min 11.1%. Clarke zone A 85.4% at 60 min, zones C to E 2.5%.

40 test patients never seen in training or calibration. Scored on days 8 to 30.

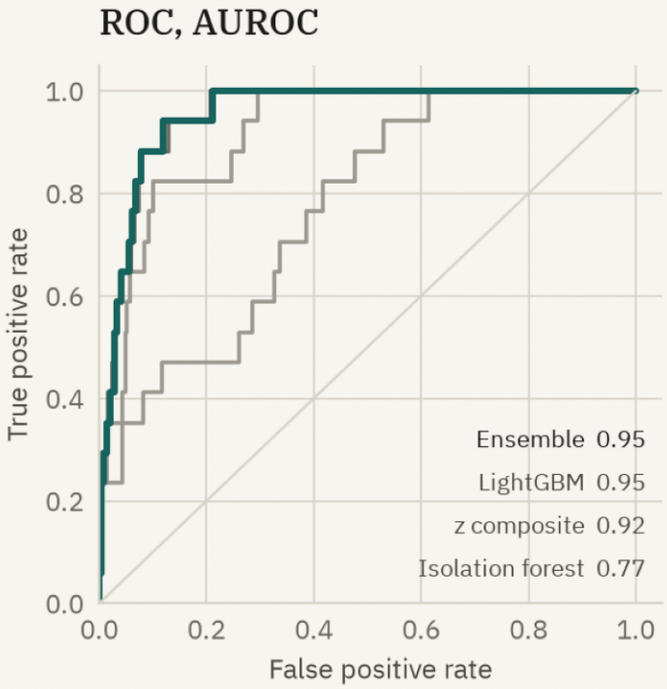
Results: alerts and honest uncertainty

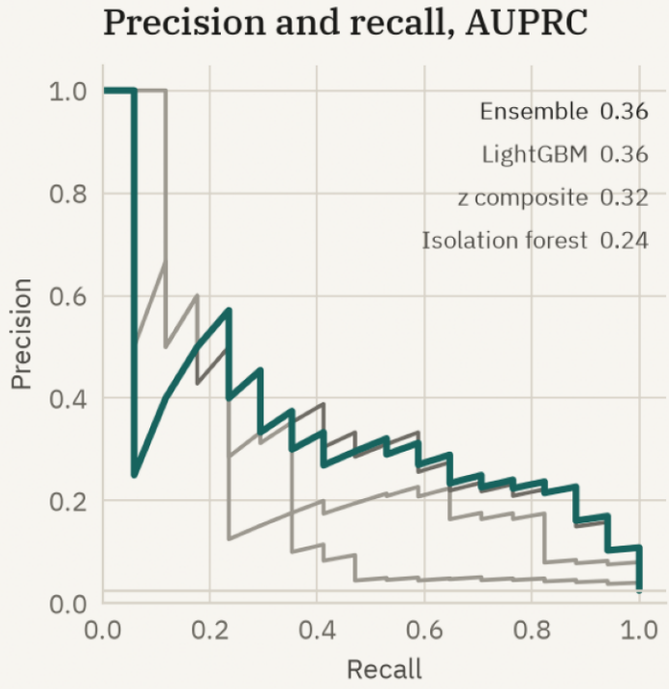


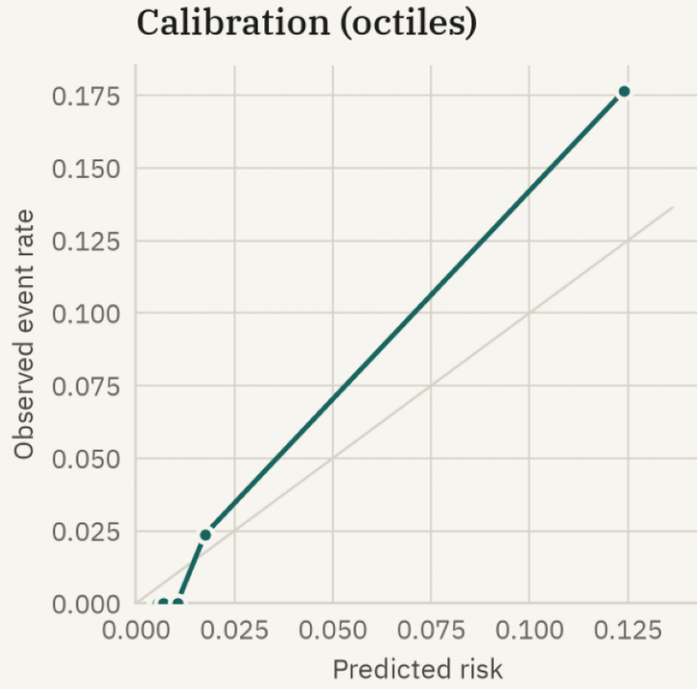


- Alert tier: 67% of spikes flagged, 85% precision, 0.20 false alarms per patient day, median 30 min ahead.
- Watch tier: 90% flagged a median 75 min ahead, shown but not pushed.
- Coverage of the 90% interval: 89.8% to 88.7% across horizons.

Results: cardiovascular risk flag

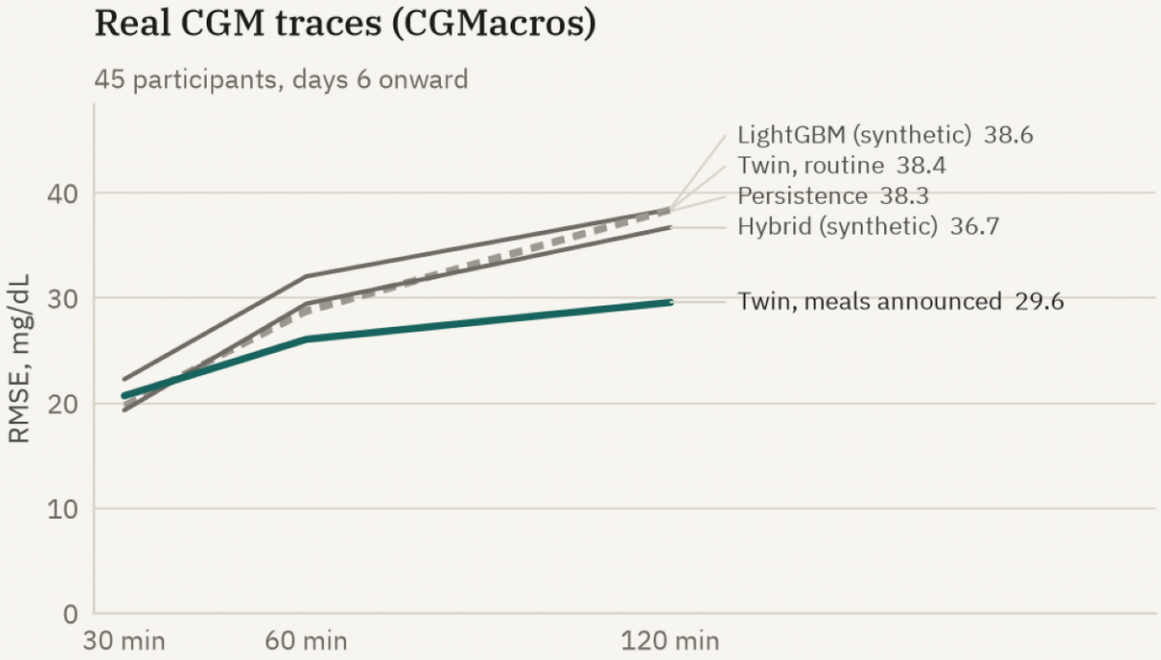


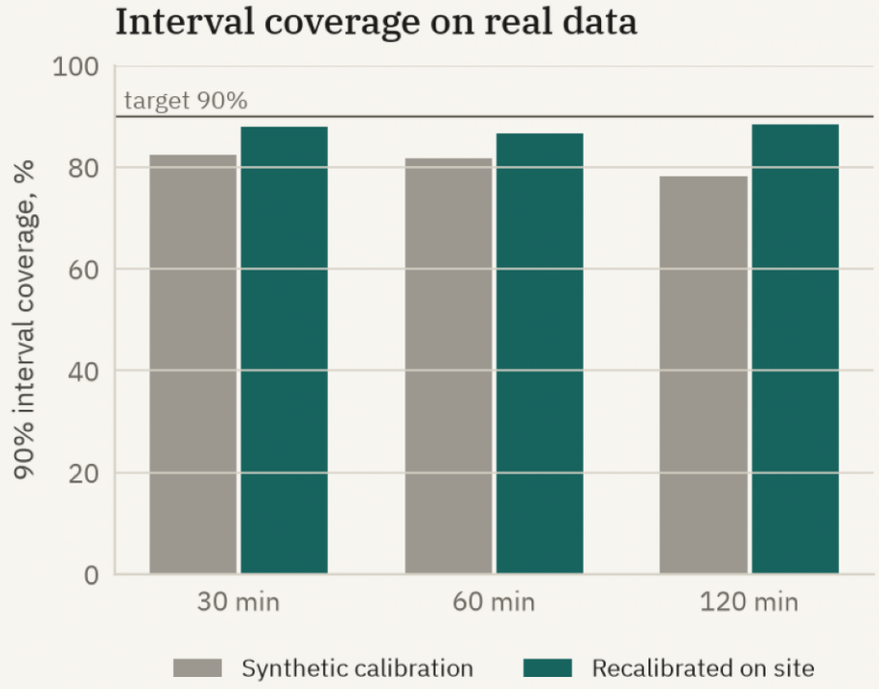




Ensemble AUROC 0.95 (95% CI 0.90 to 0.99), AUPRC 0.36 at 2.5% prevalence, sensitivity 88% at 90% specificity. At the 95% specificity operating point 5 of 6 test events were flagged up to 3 days ahead. Only 6 test events, so intervals are wide. The events come from our own simulator: with its warning signs weakened to half strength the flag catches 4 of 6, at a quarter 1 of 6. A demonstration of the pipeline, not a clinical claim.

Real CGM traces: what transfers and what does not

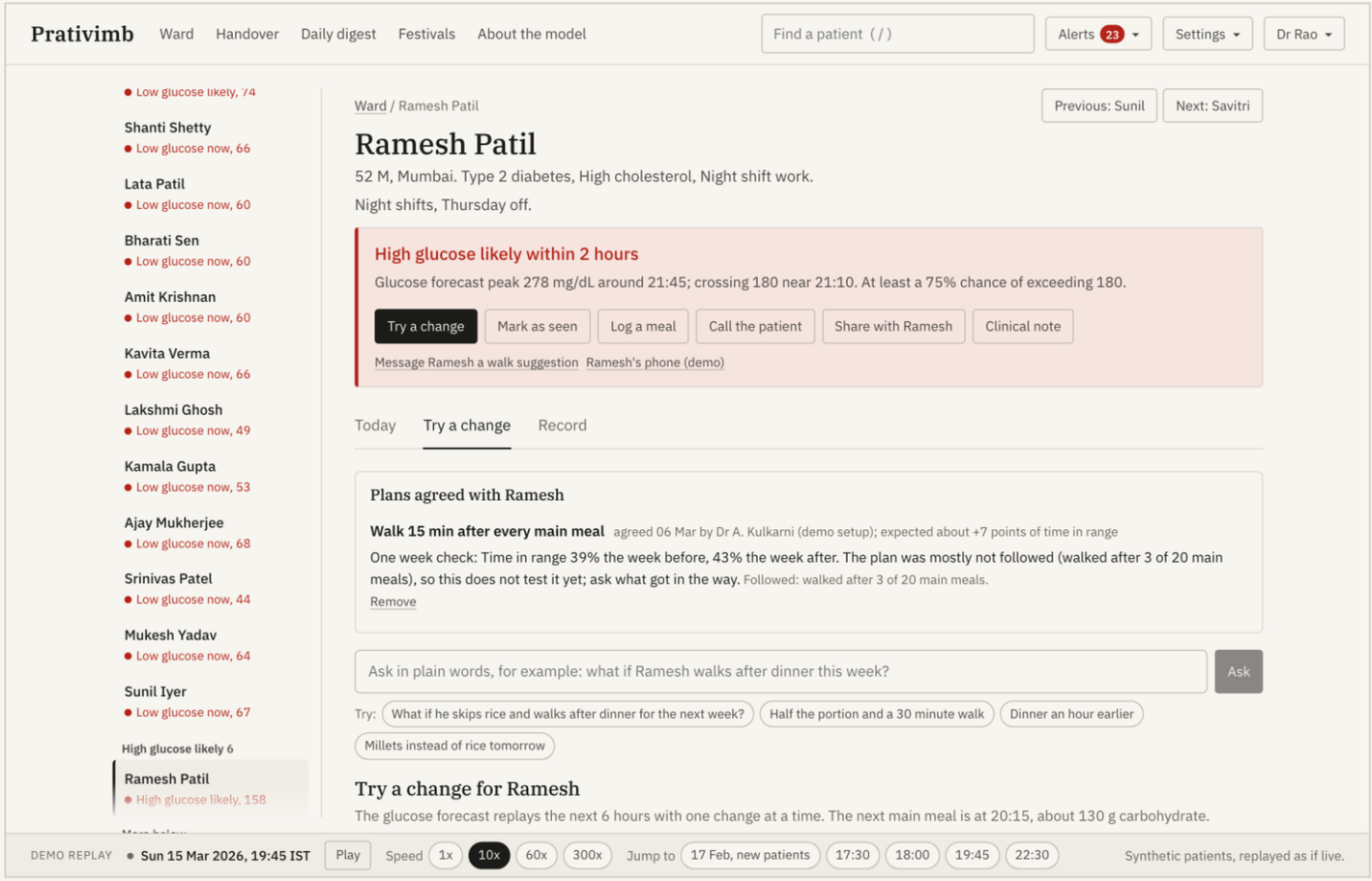




CGMacros (PhysioNet, open access): 45 adults, Dexcom G6, meals with known carbohydrate. Trained only on synthetic data, the hybrid lands near persistence (36.7 against 38.3 mg/dL at 2 h). Told about meals in advance, the calibrated twin reaches 29.6. Interval coverage recovers from 78% to 88% after per site recalibration.

CASE STUDY

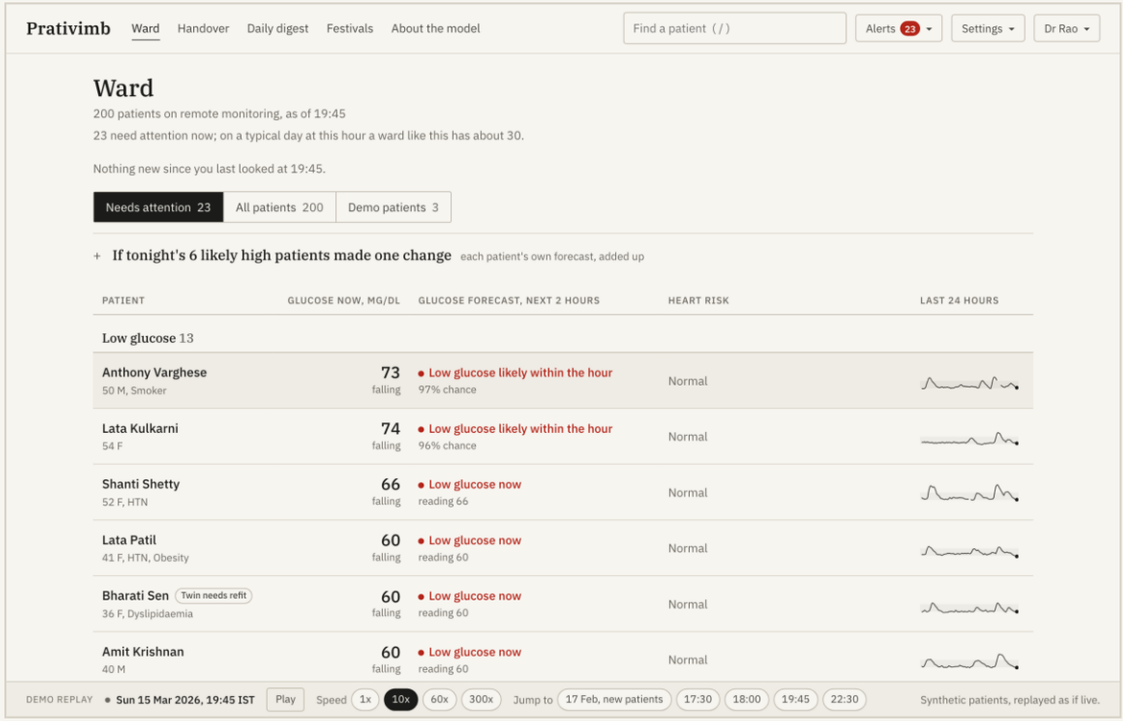
What if: Ramesh, 52, night shift, T2D



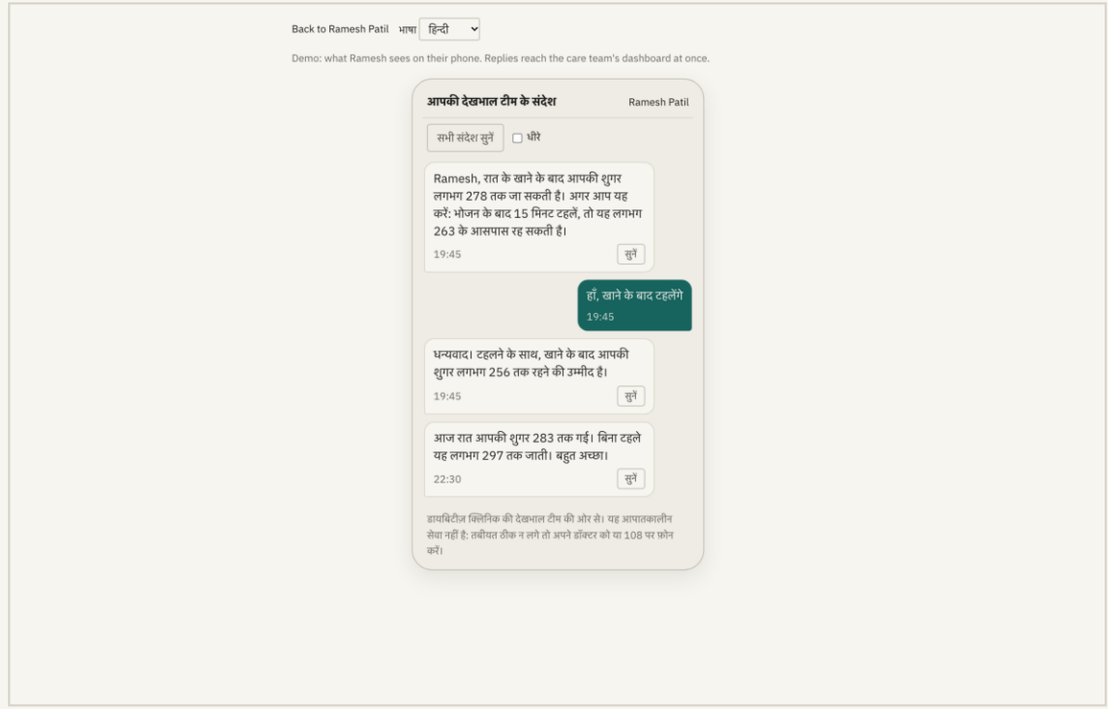
- 19:45. The hybrid forecast expects dinner at 20:15 and a peak near 278 mg/dL, after three nights of 4.9 h sleep.
- A 15 minute walk after dinner takes 15 mg/dL off the twin's 6 hour peak. Millets instead of white rice take 48.
- Together: peak 57 mg/dL lower and 40 minutes less above 180.
- Moving dinner earlier barely changes the peak tonight (it can only move to 19:50). The twin can say no.
- Checked against the reference simulator the twin never sees: across 316 moments in 40 test patients the twin gets the direction of a change right 76% of the time.

USE

Clinical workflow



1. Ward grouped by urgency, with why each alert fired; n opens the next unseen alert. Load measured: 1.3 confident high alerts per patient per day, 94% came true. 2. The first action fits the alert: check now, call, book a review, try a change. Undo for 10 seconds.



3. A suggestion to the patient's phone in Hindi, Tamil or English; the reply enters the forecast and the outcome comes back to both sides. 4. Every alert's outcome, week plans checked a week later, handover, festival planner.

GOVERNANCE

Privacy, safety and limits

- Synthetic data only: no personal data under the DPDP Act 2023 or HIPAA was used.
- Consent on screen: the patient can withdraw glucose, wearable or ABDM record sharing, and the service stops using that source.
- Roles checked by the service: doctor, nurse or health educator, patient. Every action and undo is audited.
- Deployment design: on premises inference, no identifiers in model features, clinic single sign on.
- Decision support only, not a diagnosis. The clinician decides.

Limitations we state openly

- Synthetic ground truth is still a model; real data will be harder.
- Only 6 cardiovascular events in test patients, from our own simulator; no real outcome validation yet.
- Conformal coverage is approximate with serially correlated readings, and lower below 70 mg/dL.
- Unlogged meals remain the main source of missed spikes.

NEXT

Impact and roadmap

Now	Working prototype: data, twin, models, API, dashboard, tests, docs. Reproducible from one seed.
3 months	Retrospective validation on de-identified CGM and EHR data under an ethics approval; Indian food absorption library.
6 months	Federated learning of the residual model across hospitals; the twin stays per patient, only model updates move.
9 months	ABDM integration (health ID, consent manager, FHIR); real device APIs (Libre, Dexcom, Health Connect).
12 months	Prospective pilot in a diabetes clinic with alert burden and time in range as outcomes.

Impact: fewer hours above range, earlier review of cardiac drift, and conversations about change grounded in the patient's own physiology.

Code: <public repository URL>. Demo video: <unlisted YouTube URL>. Thank you.